Exploratory Data Analysis and Machine Learning on Movie Titles Dataset

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Abstract—**This paper presents an exploratory data analysis of a movie and TV show dataset using Python. The study focuses on data cleaning, visualization, and identifying patterns in content types, genres, release trends, and ratings. A Linear Regression model is also implemented to predict IMDb scores using selected features. The results highlight key trends in content production and demonstrate the effectiveness of data analysis and machine learning techniques.**

Introduction

With the rapid growth of digital streaming platforms such as Netflix, Amazon Prime, and Disney+, the entertainment industry has experienced a massive increase in content production. This has resulted in large datasets containing valuable information about movies and television shows.

Analyzing such datasets is important for understanding audience preferences, identifying popular genres, and studying trends in content production over time. Data analysis techniques enable researchers to extract meaningful insights from raw data, while visualization helps in presenting these insights in an understandable format.

In this project, a dataset containing information about movies and TV shows is analyzed using Python. The study focuses on exploring patterns in content type, release year, genres, and ratings. Additionally, a machine learning model is developed to predict IMDb scores based on selected features. This combination of data analysis and predictive modeling demonstrates the practical application of data science techniques in real-world scenarios.

Ease of Use

The implementation of this project is straightforward due to the use of Python, which is widely recognized for its simplicity and readability. Python provides a rich ecosystem of libraries that support data analysis, visualization, and machine learning.

Pandas is used for data manipulation and preprocessing, allowing efficient handling of large datasets. NumPy supports numerical computations, while Matplotlib and Seaborn provide powerful tools for creating visualizations such as bar charts, histograms, and line plots.

Scikit-learn simplifies the implementation of machine learning algorithms by offering pre-built models and evaluation techniques. The Linear Regression model used in this project is easy to implement and interpret, making it suitable for beginners.

Overall, the tools used in this project are easy to learn and implement, making the entire workflow efficient and accessible.

DATASET DESCRIPTION

The dataset used in this project contains comprehensive information about movies and television shows collected from publicly available sources. It includes a wide range of attributes that describe different aspects of each title, enabling detailed analysis of content trends and audience preferences.

The dataset consists of both categorical and numerical features. The categorical attributes include content type (movie or TV show), genres, and production countries, which help in understanding the distribution of content across categories and regions. The numerical attributes include release year, runtime, IMDb score, TMDB score, TMDB popularity, and IMDb votes, which are useful for statistical analysis and machine learning.

The **type** column distinguishes between movies and television shows, allowing comparison between different forms of content. The **release_year** attribute represents the year in which the title was released, which is useful for analyzing trends over time. The **runtime** feature indicates the duration of the content, providing insight into variations in content length.

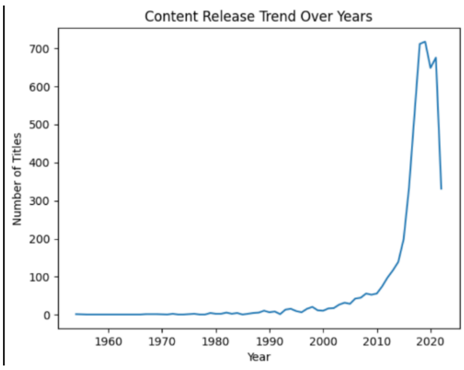
The **genres** column contains information about the category of each title, such as Drama, Comedy, or Action. This helps in identifying the most popular types of content. The **production_countries** attribute indicates the origin of the content, which allows analysis of global production patterns.

The **IMDb score** and **TMDB score** represent ratings provided by users on different platforms, while **TMDB popularity** reflects how popular a title is based on user interactions. The **IMDb votes** feature indicates the number of users who rated a particular title, which adds reliability to the rating.

Before performing analysis, the dataset was preprocessed to ensure quality and consistency. Missing values in critical columns such as IMDb score and release year were removed, while less important missing values were handled using appropriate techniques such as filling with median values or assigning default labels. Duplicate entries were also removed to maintain data integrity.

Additionally, only relevant columns were selected to reduce complexity and improve computational efficiency. This step ensured that the dataset was clean, well-structured, and suitable for exploratory data analysis and machine learning.

Overall, the dataset provides a rich source of information for analyzing trends in the entertainment industry and supports the development of predictive models.



METHODOLOGY

A. Data Cleaning

Data cleaning is a crucial step in the analysis process. Missing values in key columns such as IMDb score and release year were removed to ensure accuracy. Other missing values, such as runtime and genres, were filled using appropriate techniques. Duplicate entries were also identified and removed.

B. Exploratory Data Analysis (EDA)

EDA was performed to gain insights into the dataset. The analysis included:

- Distribution of movies and TV shows
- Year-wise trend of content production
- Identification of most frequent genres
- Analysis of production countries
- Distribution of ratings and popularity

EDA helps in understanding the structure of the data and identifying patterns before applying machine learning models.

C. Data Visualization

Visualization plays an important role in data analysis. Various plots were created to represent the data visually. Bar charts were used to compare categories such as content type and genres. Line graphs were used to analyze trends over time, particularly the growth in content production.

Histograms were used to study the distribution of IMDb scores and TMDB popularity. Additionally, a heatmap was used to visualize correlations between numerical features, helping to identify relationships between variables.

These visualizations made it easier to interpret the data and support the findings of the analysis.

D. Machine Learning Model

A Linear Regression model was implemented to predict IMDb scores based on selected features, including release year, runtime, TMDB popularity, and TMDB score.

The dataset was split into training and testing sets to evaluate the model's performance. Proper preprocessing ensured that the data was suitable for modeling.

The model was evaluated using Mean Squared Error (MSE) and R^2 score. These metrics helped in understanding how well the model fits the data and predicts the target variable.

RESULTS AND DISCUSSION

The analysis provided several important insights into the dataset.

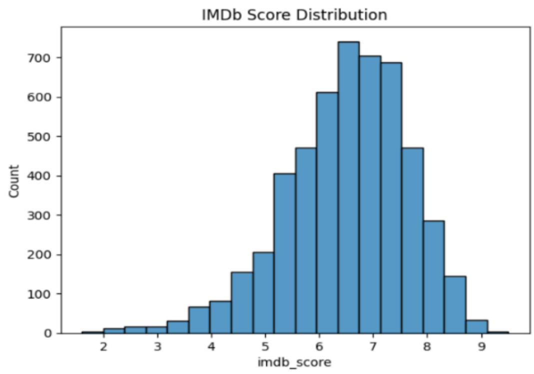
The comparison between movies and TV shows shows that movies are significantly more common, indicating a higher production rate in the film industry. The analysis of release years reveals a clear upward trend in content production, especially after the year 2000, reflecting the growth of digital platforms.

Genre analysis shows that categories such as Drama, Comedy, and Action dominate the dataset, suggesting strong audience preference for these genres. Production country analysis indicates that a few countries contribute the majority of content, highlighting an imbalance in global content production.

The distribution of IMDb scores shows that most titles fall within the range of 6 to 8, indicating generally favorable ratings. Popularity analysis reveals that while a small number of titles achieve very high popularity, most remain moderately popular.

The correlation heatmap highlights relationships between numerical features. A strong positive correlation between IMDb score and TMDB score indicates consistency between different rating platforms. Other features, such as runtime, show weaker relationships with ratings.

The machine learning model demonstrated moderate performance in predicting IMDb scores. This suggests that while the selected features contribute to prediction, additional factors may influence ratings.

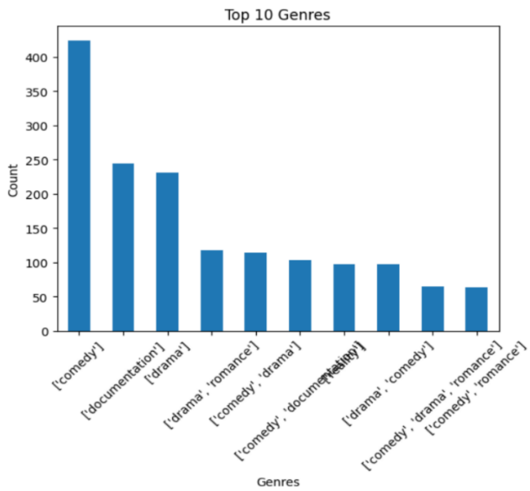


CONCLUSION

This project demonstrates the application of data analysis and machine learning techniques to a real-world dataset. The analysis successfully identified trends in content production, genre distribution, and ratings.

Visualization techniques played a key role in understanding the data, while the machine learning model provided predictive insights. The results highlight the usefulness of Python-based tools in performing data analysis tasks efficiently.

Overall, this study shows how data-driven approaches can be used to gain meaningful insights and support decision-making.



FUTURE SCOPE

The current study focuses on basic data analysis and a simple machine learning model. Future work can involve the use of advanced algorithms such as Decision Trees, Random Forest, and Neural Networks to improve prediction accuracy.

Additionally, natural language processing techniques can be applied to analyze textual data such as movie descriptions and reviews. Expanding the dataset and incorporating additional features can further enhance the analysis.

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